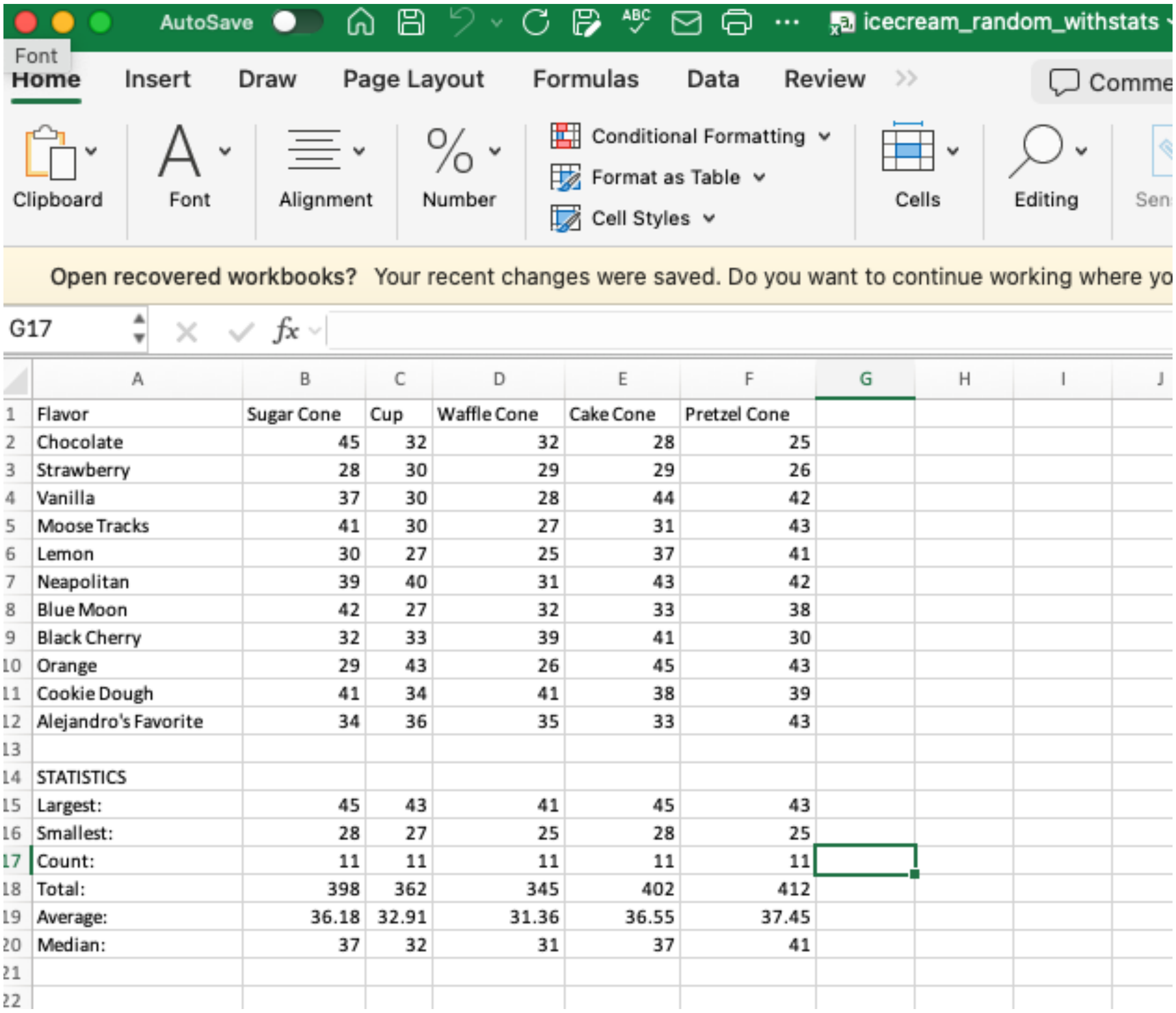
Aejandro De La Torre
LIS 440
Lecture: *Statistical Calculations in Pandas: Randomized Ice Cream Data*
11/18/25



	A	B	C	D	E	F
1	Flavor	Sugar Cone	Cup	Waffle Cone	Cake Cone	Pretzel Cone
2	Chocolate	45	32	32	28	25
3	Strawberry	28	30	29	29	26
4	Vanilla	37	30	28	44	42
5	Moose Tracks	41	30	27	31	43
6	Lemon	30	27	25	37	41
7	Neapolitan	39	40	31	43	42
8	Blue Moon	42	27	32	33	38
9	Black Cherry	32	33	39	41	30
10	Orange	29	43	26	45	43
11	Cookie Dough	41	34	41	38	39
12	Alejandro's Favorite	34	36	35	33	43
13						
14	STATISTICS					
15	Largest:	45	43	41	45	43
16	Smallest:	28	27	25	28	25
17	Count:	11	11	11	11	11
18	Total:	398	362	345	402	412
19	Average:	36.18	32.91	31.36	36.55	37.45
20	Median:	37	32	31	37	41
21						
22						

Code

```
# Import the necessary packages
import pandas as pd
import os
from pathlib import Path

# Print version of pandas just for tracking purposes
print(pd.__version__)

# Because the professor does this
print("Alejandro De La Torre's Code...")

# Set current working directory (CWD) : root folder
root = Path(__file__).parent.parent.resolve()

# Show CWD
print("Current Working Directory: ",Path.cwd())

# Set CWD: Data
```

```
data = root / 'data'
# Set CWD: Data > Raw
data_raw = data / 'raw'
# Set CWD: Data > Processed
data_processed = root / 'data' / 'processed'
# Set CWD: Scripts
scripts = root / 'scripts'


# Read Randomized Ice Cream Data
icecream_data = data_processed / 'icecream_random.csv'

# Create dataframe of Ice Cream (IC) data
ic_df = pd.read_csv(icecream_data) # He labeled it wisconsinHousingDF but too long


# View first and last 3 entries for vibes
print("First 3 rows: \n", ic_df.head(3))
print("Last 3 rows: \n", ic_df.tail(3))


# Show and select the sugar column
sugar_col = ic_df['Sugar Cone']
print("Sugar Cone Sales:") # to print raw series data
print(sugar_col)

# Calculate total
print("Total Sugar Cone Sales:")
print(sugar_col.sum())

# Calculate largest (max)
print("Largest Sugar Cone Sales:")
print(sugar_col.max())

# Calculate smallest (min)
print("Smallest Sugar Cone Sales:") # Least? Minimum? Why smallest/largest?
print(sugar_col.min())

# Calculate count (of entries)
print("Number of Sugar Cone Sales Data Points:")
print(sugar_col.count())

# Calculate median
print("Median Sugar Cone Sales for All Flavors:")
print(sugar_col.median())

# Calculate average (mean)
print("Average Sugar Cone Sales for All Flavors:")
print(sugar_col.mean())
```

Shell

```
>>> %Run pandas_icecream_stats.py
2.3.3
Alejandro De La Torre's Code...
```

```
Current Working Directory:  /Users/alexdelatorre/Library/CloudStorage/OneDrive-UW-Madison/Coursework/L I S 440/Lectures/Week 12/Tuesday/scripts
First 3 rows:
      Flavor  Sugar Cone  Cup  Waffle Cone  Cake Cone  Pretzel Cone
0   Chocolate          45   32           32         28           25
1  Strawberry          28   30           29         29           26
2     Vanilla          37   30           28         44           42
Last 3 rows:
                  Flavor  Sugar Cone  Cup  Waffle Cone  Cake Cone  Pretzel Cone
8                 Orange          29   43           26         45           43
9           Cookie Dough          41   34           41         38           39
10  Alejandro's Favorite          34   36           35         33           43
Sugar Cone Sales:
0     45
1     28
2     37
3     41
4     30
5     39
6     42
7     32
8     29
9     41
10    34
Name: Sugar Cone, dtype: int64
Total Sugar Cone Sales:
398
Largest Sugar Cone Sales:
45
Smallest Sugar Cone Sales:
28
Number of Sugar Cone Sales Data Points:
11
Median Sugar Cone Sales for All Flavors:
37.0
Average Sugar Cone Sales for All Flavors:
36.18181818181818
>>>
```